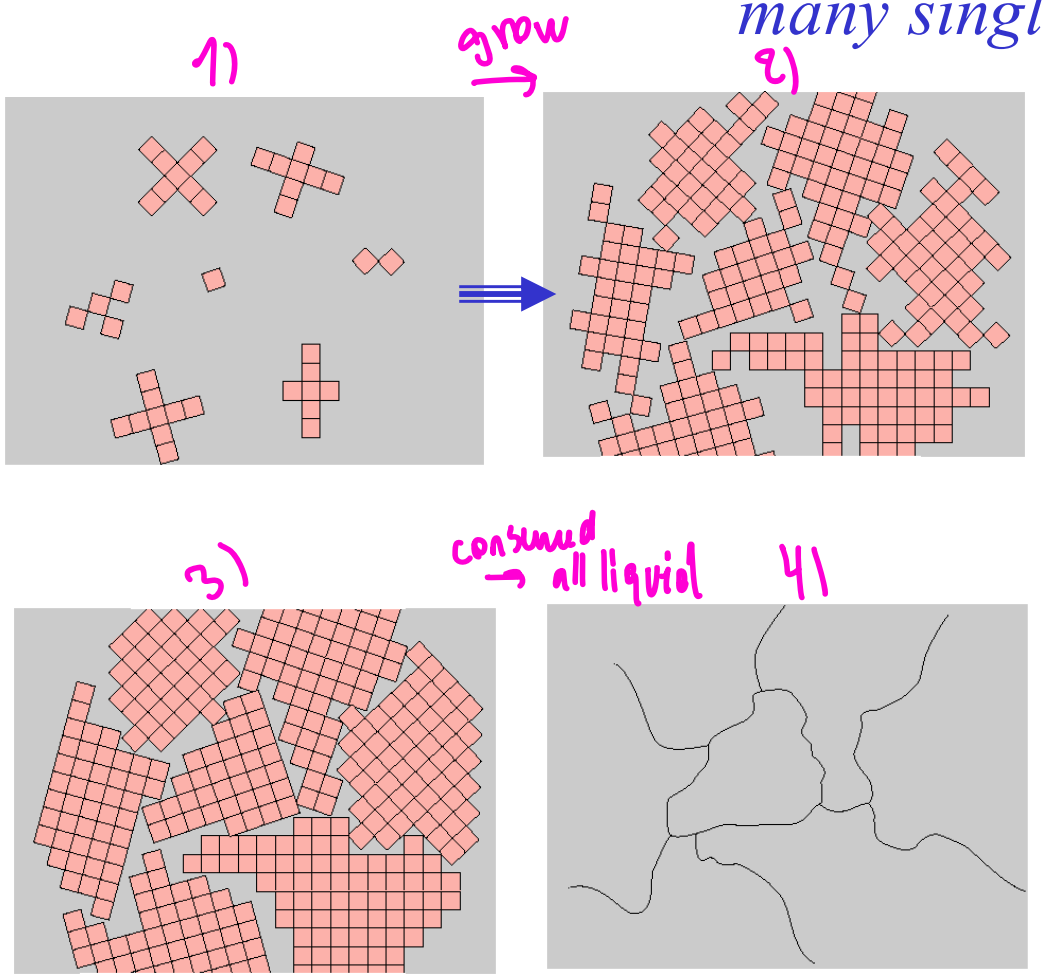
Outline:

- Crystallization
- Grain boundaries
- Grain size determination
- Stacking faults
- Types of microscopes
  - *LOM*
  - *SEM*
  - *TEM*
  - *SPM*

Polycrystalline Materials

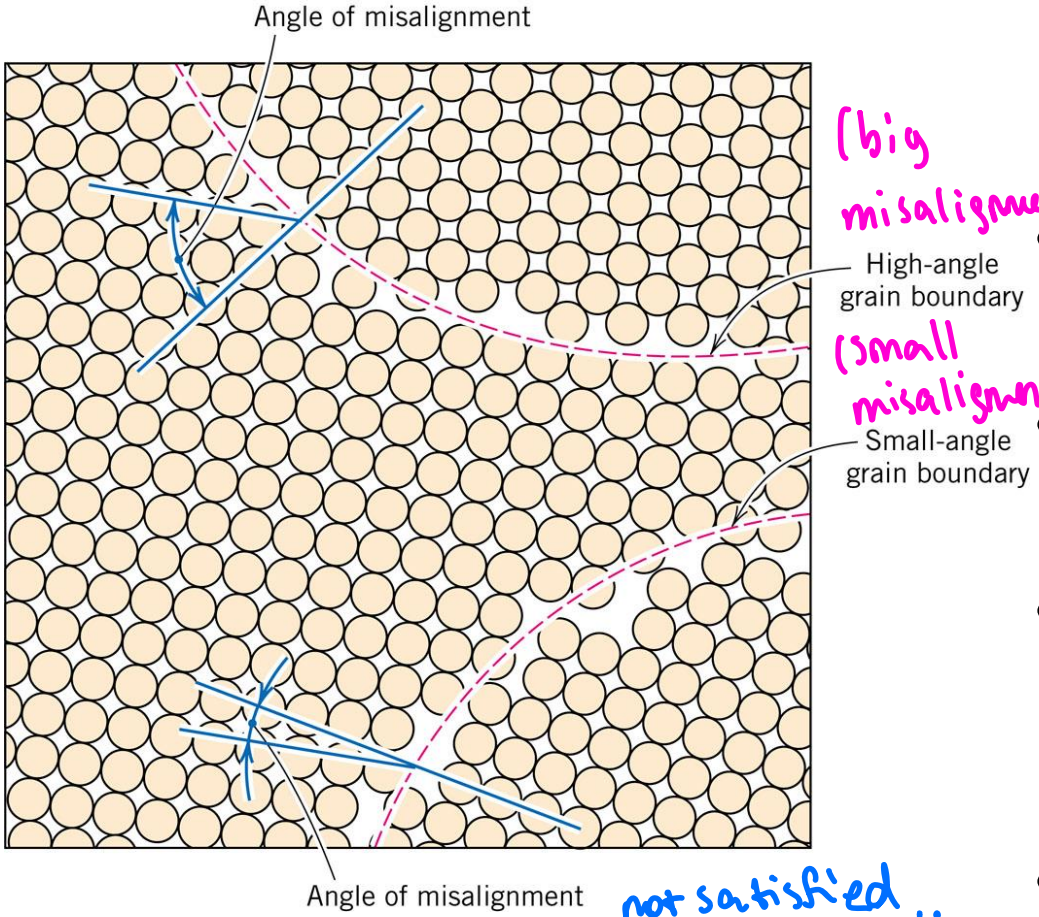
Most materials are polycrystalline and are made of many single crystals



- during *solidification* the crystals nucleate and grow from the liquid in a *random orientation*
- the *grains* impinge on each other when the solidification is complete
- junction of grains are grain boundaries

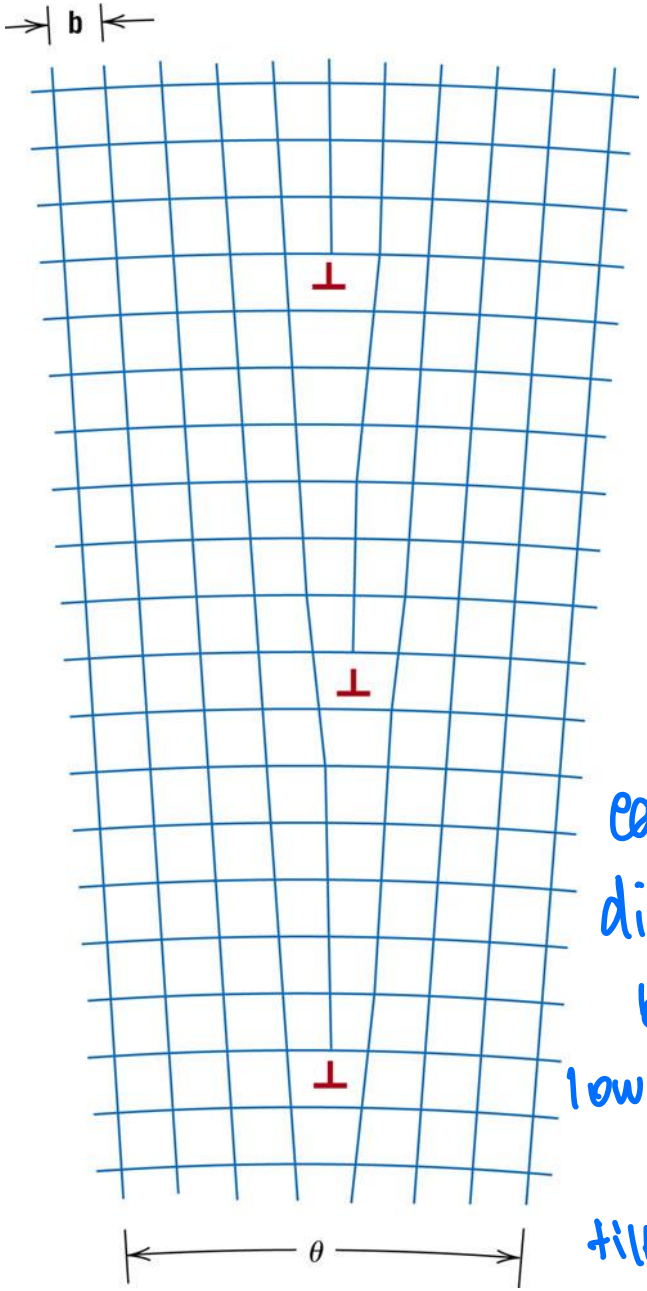
you can increase grain boundaries by making the grain smaller

Grain Boundaries



- Occurs due to the crystallographic mismatch when two grains meet
- when mis-orientation is large → ***high angle*** *grain boundary*
- when mis-orientation is small, → ***low angle*** *grain boundary*
- atoms are *less bonded* and the atomic packing is lower than in the grain (*lower coordination*)
- the result is an energy difference → *interfacial surface energy or grain boundary energy*

Grain Boundaries



edge dislocation but w/ low angle grain ⇓ tilt boundary (edge)

- grain boundaries are more chemically reactive
- segregation of impurities due to higher energy
- total grain boundary area smaller in *coarse grained* than *fine grained* material
- low angle grain boundary is described as an array of dislocations
  - *tilt boundary* (edge)
  - *twist boundary* (screw)

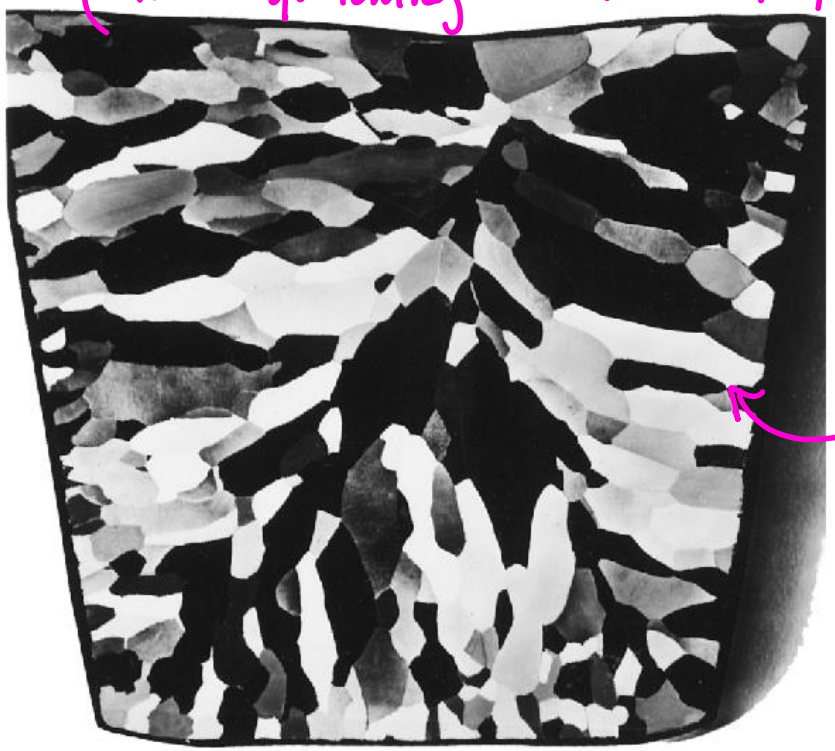
Smaller the grains ⇒ bigger the surface area

like grains of sand in jar

slow = big

Observation of Grain Structure

cooling fast = small grain
(like quenching sword in water)

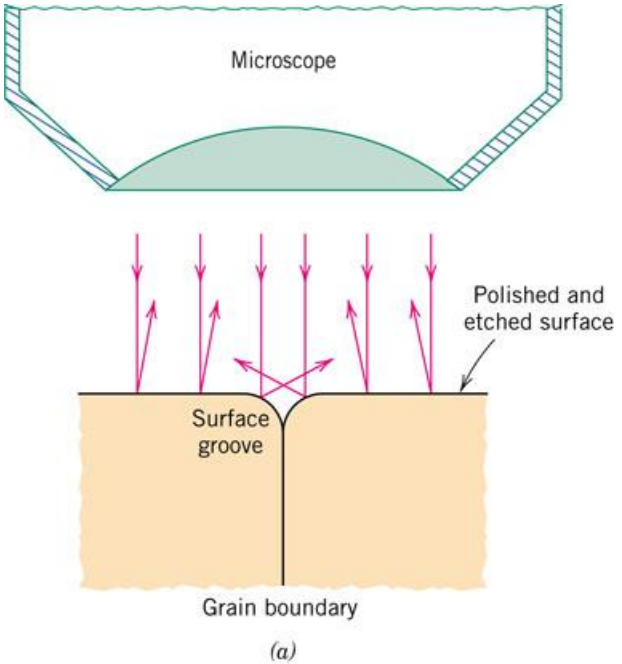


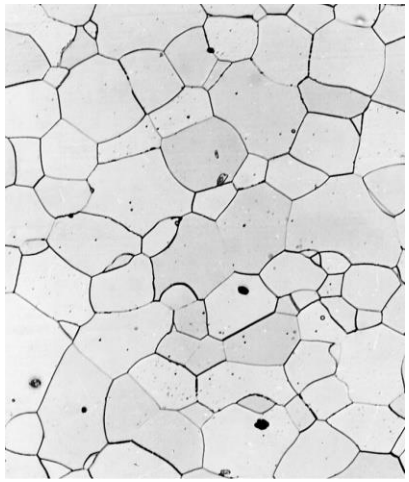
ex of slow cooling

FIGURE 4.10 *High-purity polycrystalline lead ingot in which the individual grains may be discerned.*

- *Macrostructure* can be observed with naked eye
- coarse grains can be revealed this way (e.g. Al streetlight posts, and zinc galvanized garbage cans)
- *microstructure* ... is when the grains can only be observed with a microscope
- imaged using a camera for archiving
  → *photomicrograph*

Sample Preparation for Microscopy





→ otherwise you only see scratches, dirt, fingerprints, ect

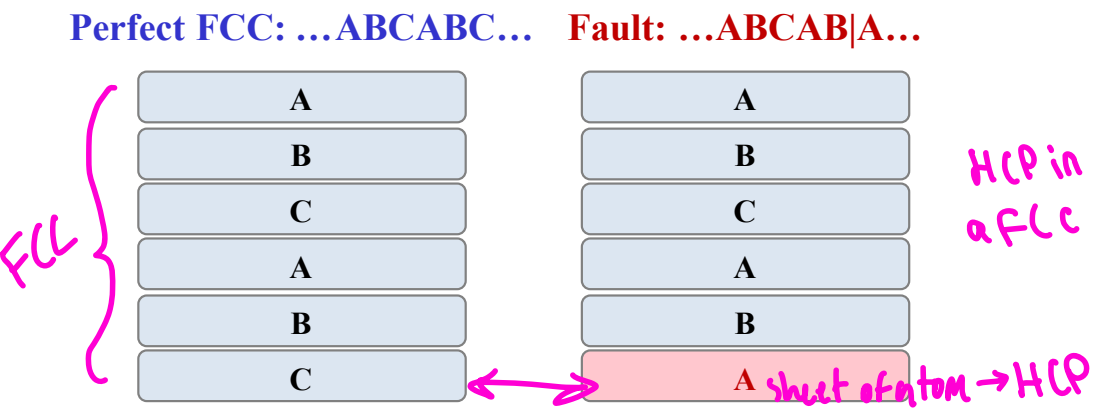
- Preparation requires meticulous *grinding* and *polishing* of the surface
- the microstructure is revealed by attack using …… (chemical reagents)
  - preferential attack of grain boundaries
- effect is that these features scatter the incident light and create **optical contrast**

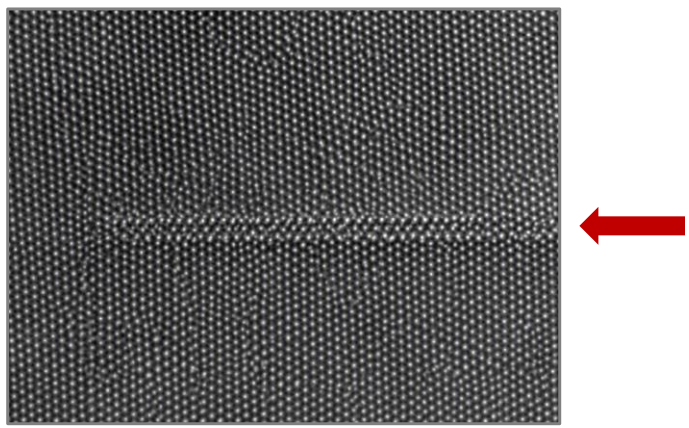
Stacking Faults

- An **area (planar) defect** – a mistake in the stacking sequence of atomic planes.
- Recall: FCC stacks as **ABCABCABC…** and HCP stacks as **ABABAB…**
- A stacking fault is when the order is broken, e.g. **ABCAB**ABC… – one plane (C) is missing.
- The atoms are still in proper positions; only the **order of the planes** is wrong, so a thin HCP-like layer appears inside the FCC crystal.

*Because the mistake is small, its energy is low – stacking faults are common. They form during **solidification and plastic deformation** and affect how easily a metal deforms.*

Compare the stacking sequences:

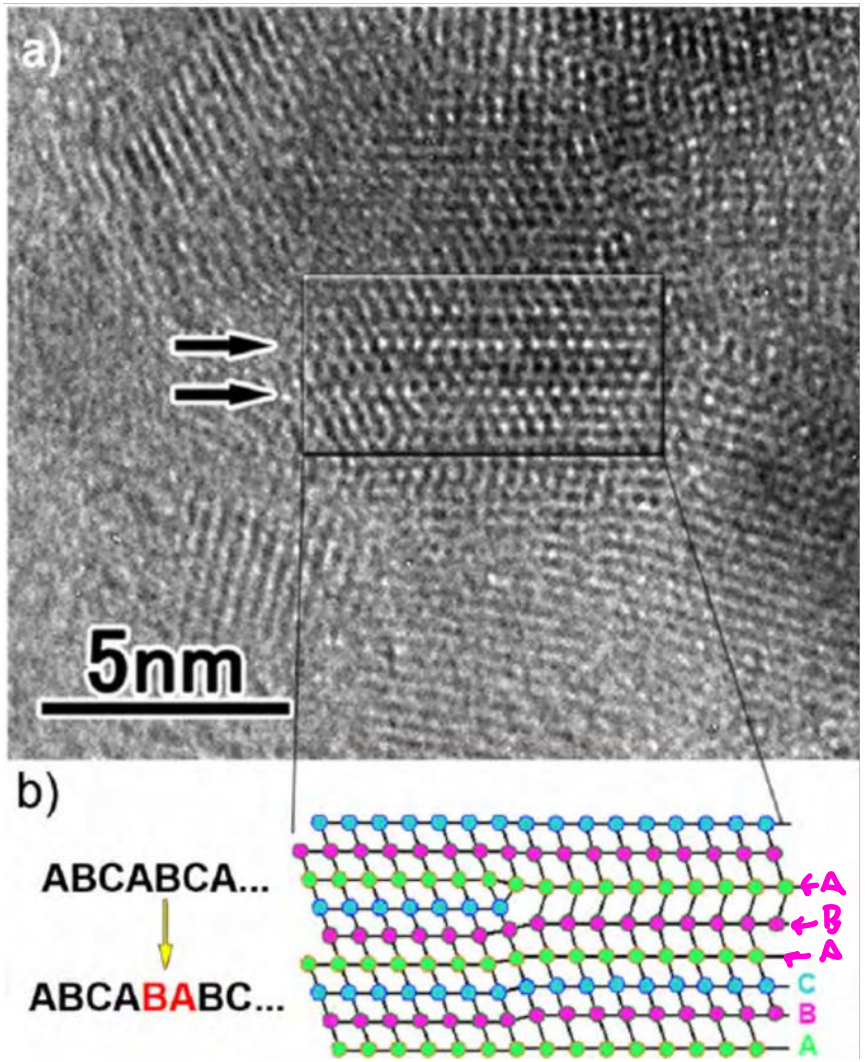




High-resolution TEM: the faulted plane appears as a break in the regular stacking

Stacking Faults

Wurtzite-type ZnS nanoparticles by pulsed electric discharge

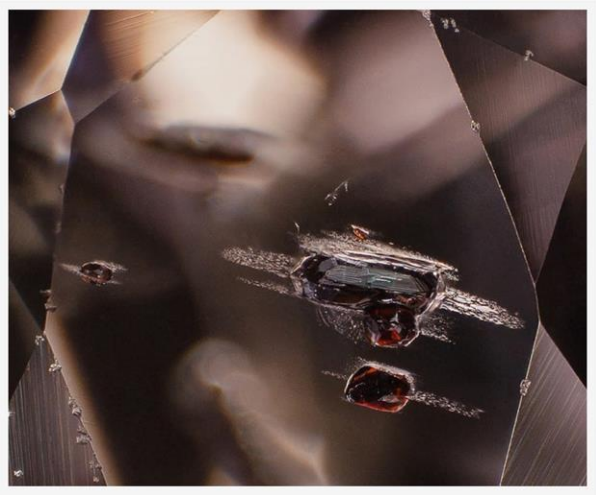


Omurzak et al. (2011) Nanotechnology.

HRTEM image of a stacking fault (a), denoted by the arrows, and (b) schematic illustration of the stacking fault in the crystal shown in (a).

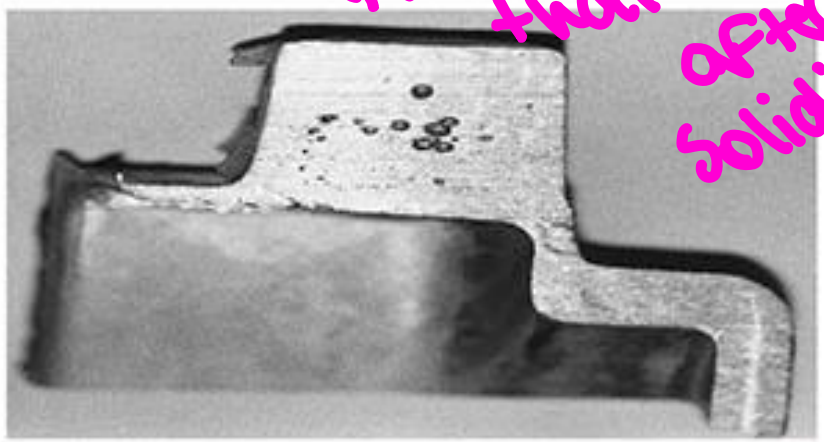
Three-dimensional (*volumetric*) defects

- Inclusions
- precipitates
- Porosities or voids
- Cracks

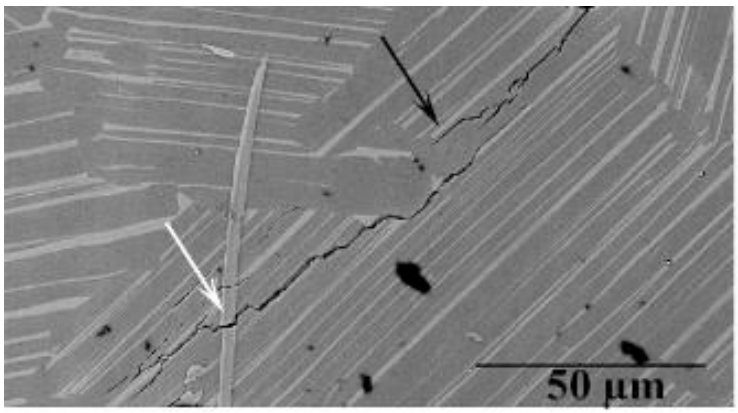


Sapphire crystal containing Rutile inclusions

trapped gas that stayed after solidification



Gas porosity in metallic component



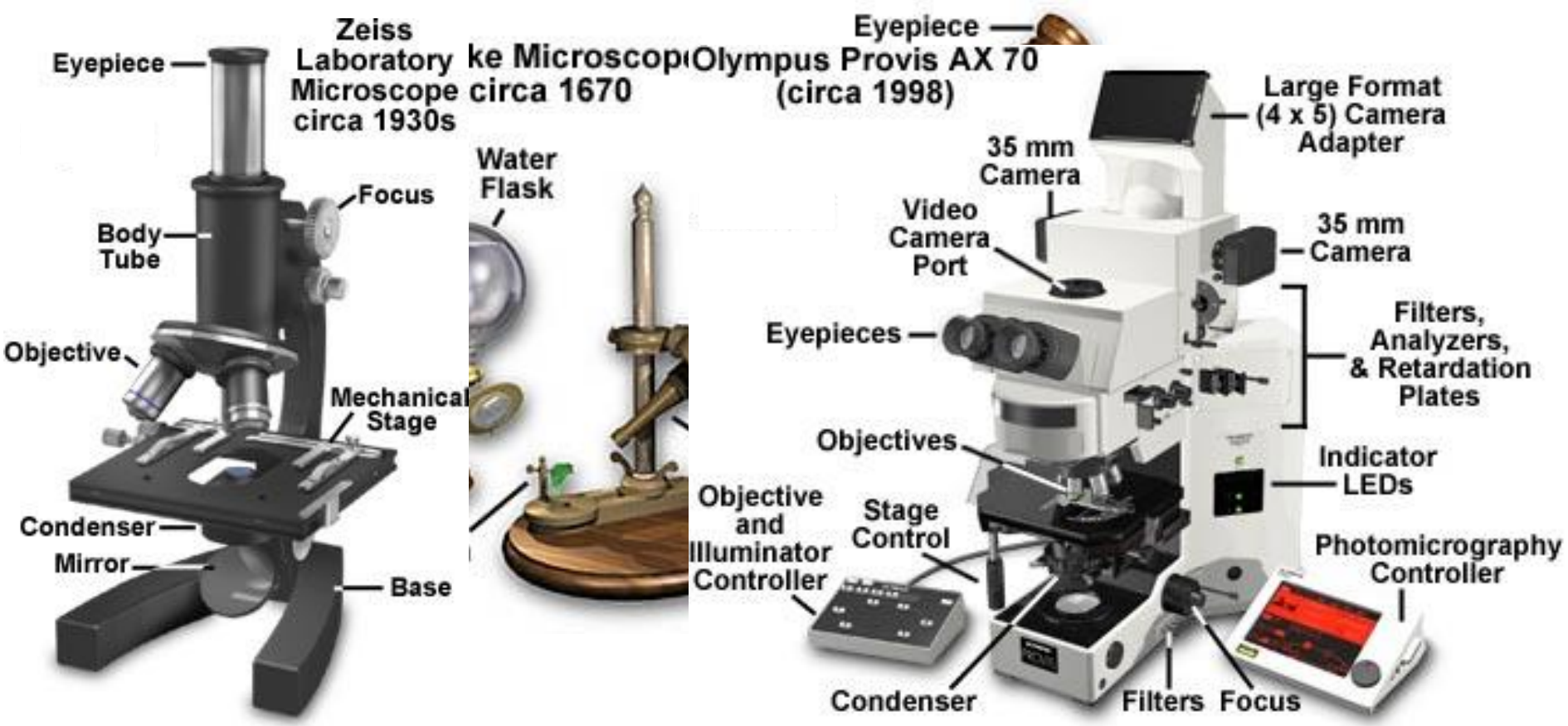
Cracks in titanium aluminide

Types of Microscopy



- *Optical microscopy is limited to x2000 magnification*
- *electron microscopy* uses electron waves (very small $\lambda \sim 0.003$nm) rather than light ($\lambda \sim 500$nm).
- can reveal microstructural features down to atomic scale (x1,000,000)
- scanning electron microscope (…….)
  - sample preparation similar to optical microscopy
  - can be used to observe fracture surface (*fractography*)
- transmission electron microscope (……..)
  - samples are very small
  - requires very thin (*electron transparent*) samples

Optical Microscopy



Review Optical Microscopy

A polished metal sample is placed under an optical microscope, but no grains can be seen — the surface just looks like a uniform bright mirror. After the sample is etched, the grains and their boundaries become clearly visible. What is the best explanation? etching attacks grain boundaries

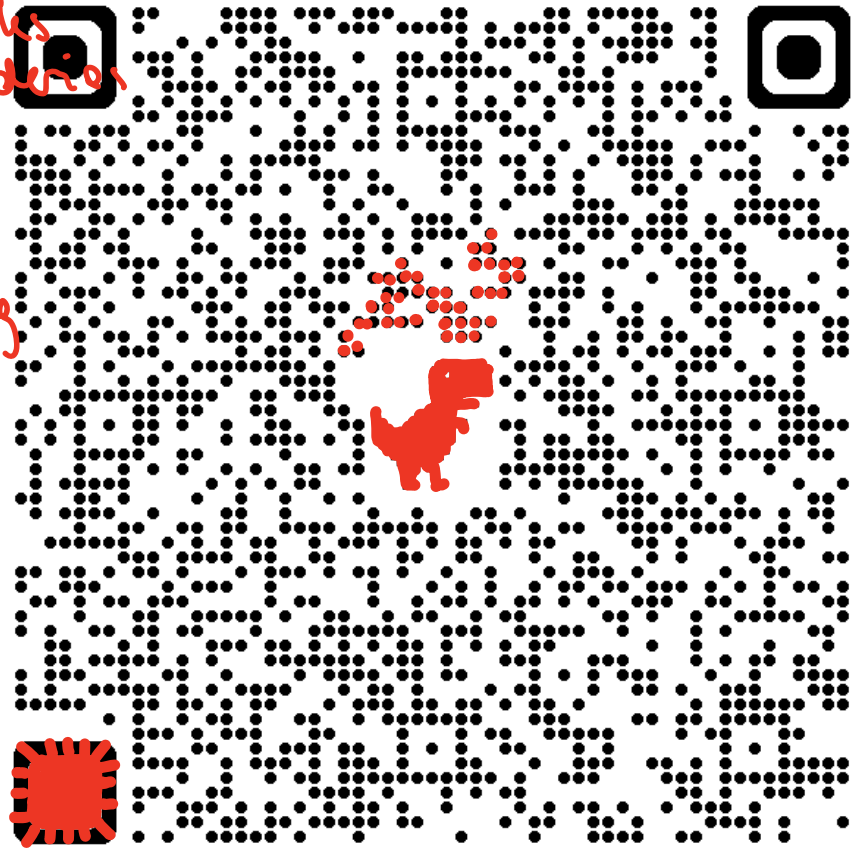
A. Etching increases the magnification so that the grains become large enough to see.

↳ colour doesn't come from etching

B. Etching colours each grain with a different chemical dye, and the optical microscope detects the colours.

C. Etching attacks grain boundaries preferentially, so the surface is no longer uniform and these features scatter the incident light differently, creating contrast.

D. Before etching the sample was a single crystal.



Use the QR code shown in the classroom. This one won't work.

Scanning Electron Microscope (SEM)



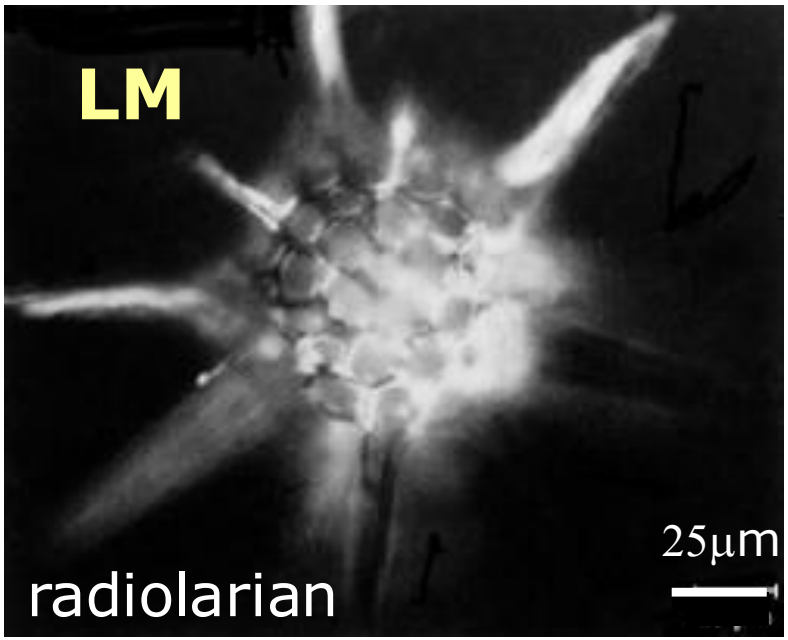
The SEM is designed for direct studying of the surfaces of solid objects

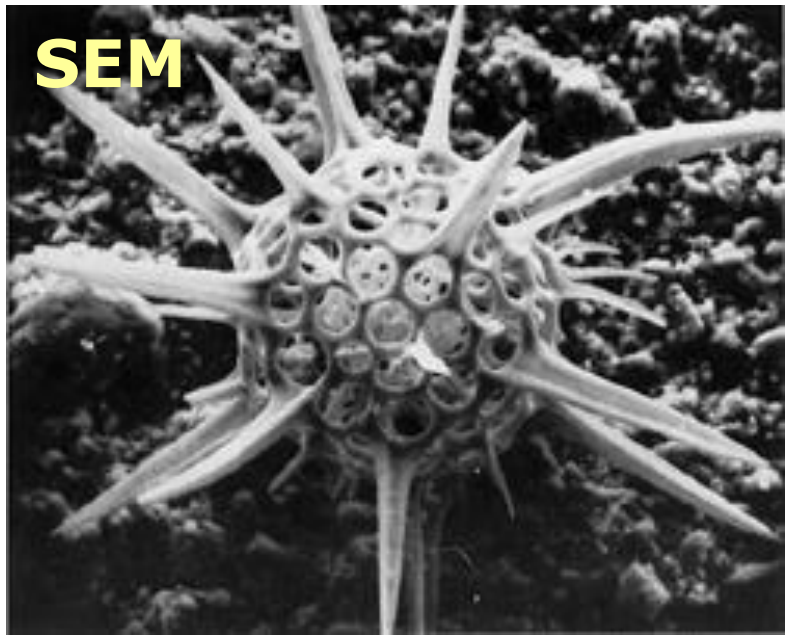
Scanning electron microscope (SEM) is a microscope that uses electrons rather than light to form an image. There are many advantages to using the SEM instead of an OM.

Advantages of Using SEM over LM

- **The SEM has a large depth of field,**

  This allows a large amount of the sample to be in focus at one time and produces an image that is a good representation of the three-dimensional sample.



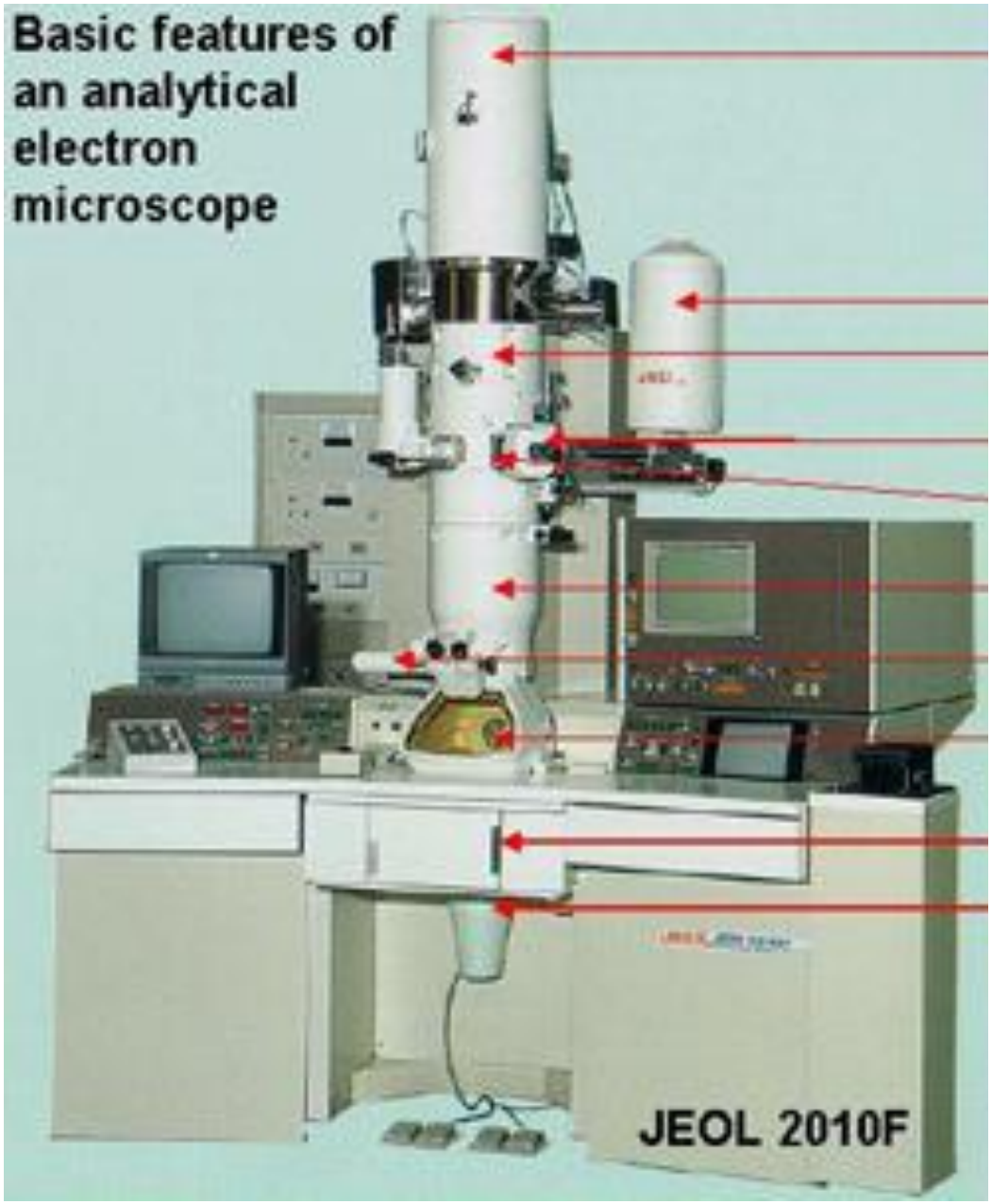


↑FOV
=
↑resolution
topogr
aphy

- **The SEM also produces images of high resolution;** ***fine features can be closely examined at a high magnification.***

- ***The combination of higher magnification, larger depth of field, greater resolution makes the SEM one of the most heavily used instruments in research areas and industries, especially in semiconductor industry.***

Transition Electron Microscope (TEM)



Electron Gun

EDS Detector
Condenser Lens

Specimen Holder
Objective Lens

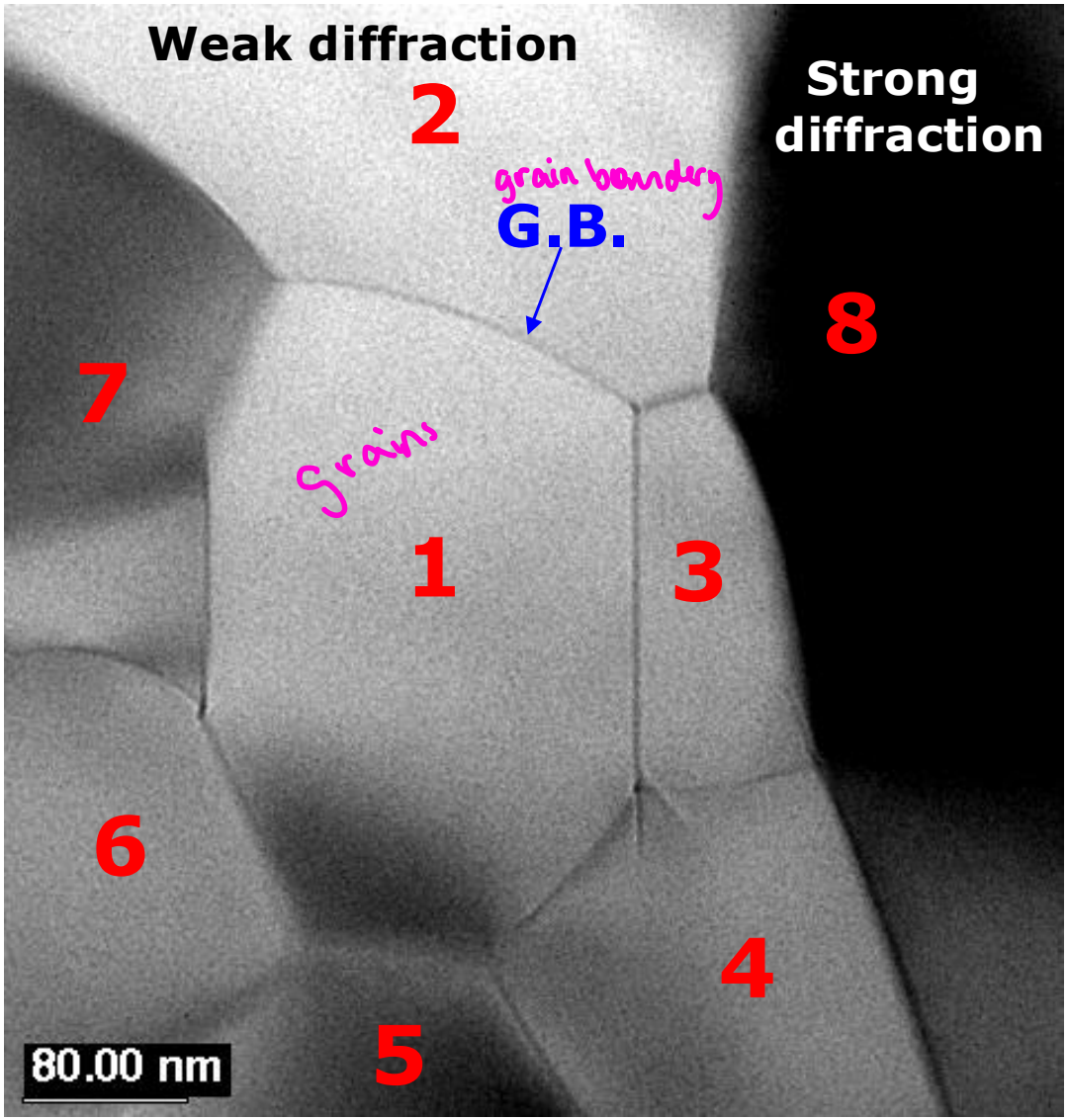
Magnifying Lenses

HAADF Detector
Viewing Chamber

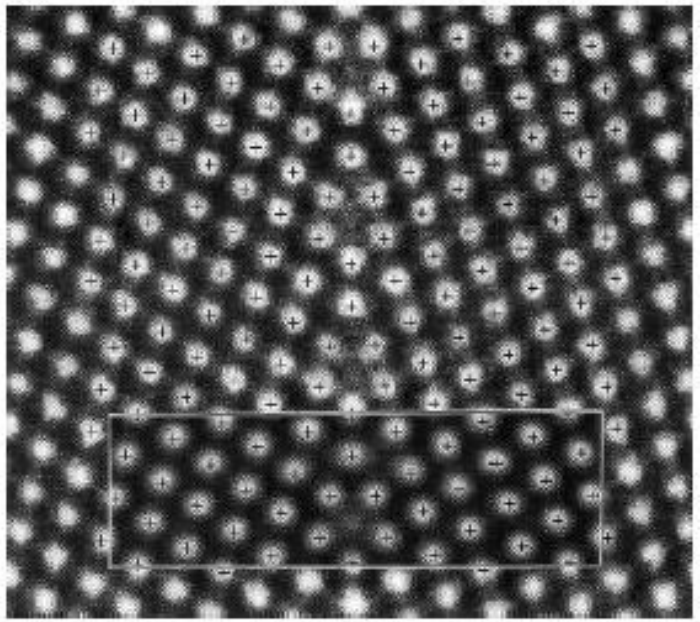
Camera Chamber

STEM Detector or EELS

Transition Electron Microscope (TEM)



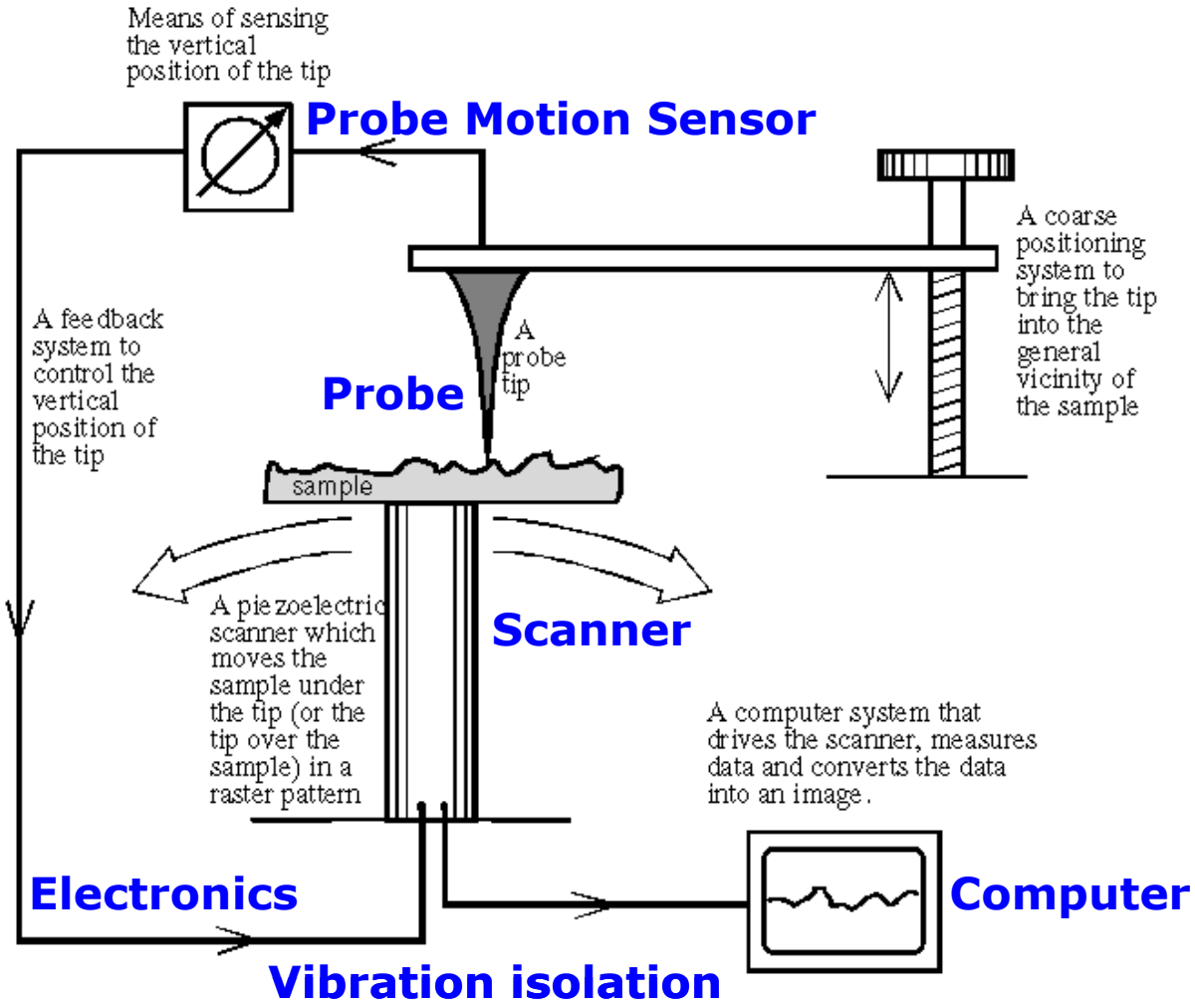
8 grains are in different orientations



High-resolution TEM image of a tilt grain boundary in Al, Sandia National Lab.

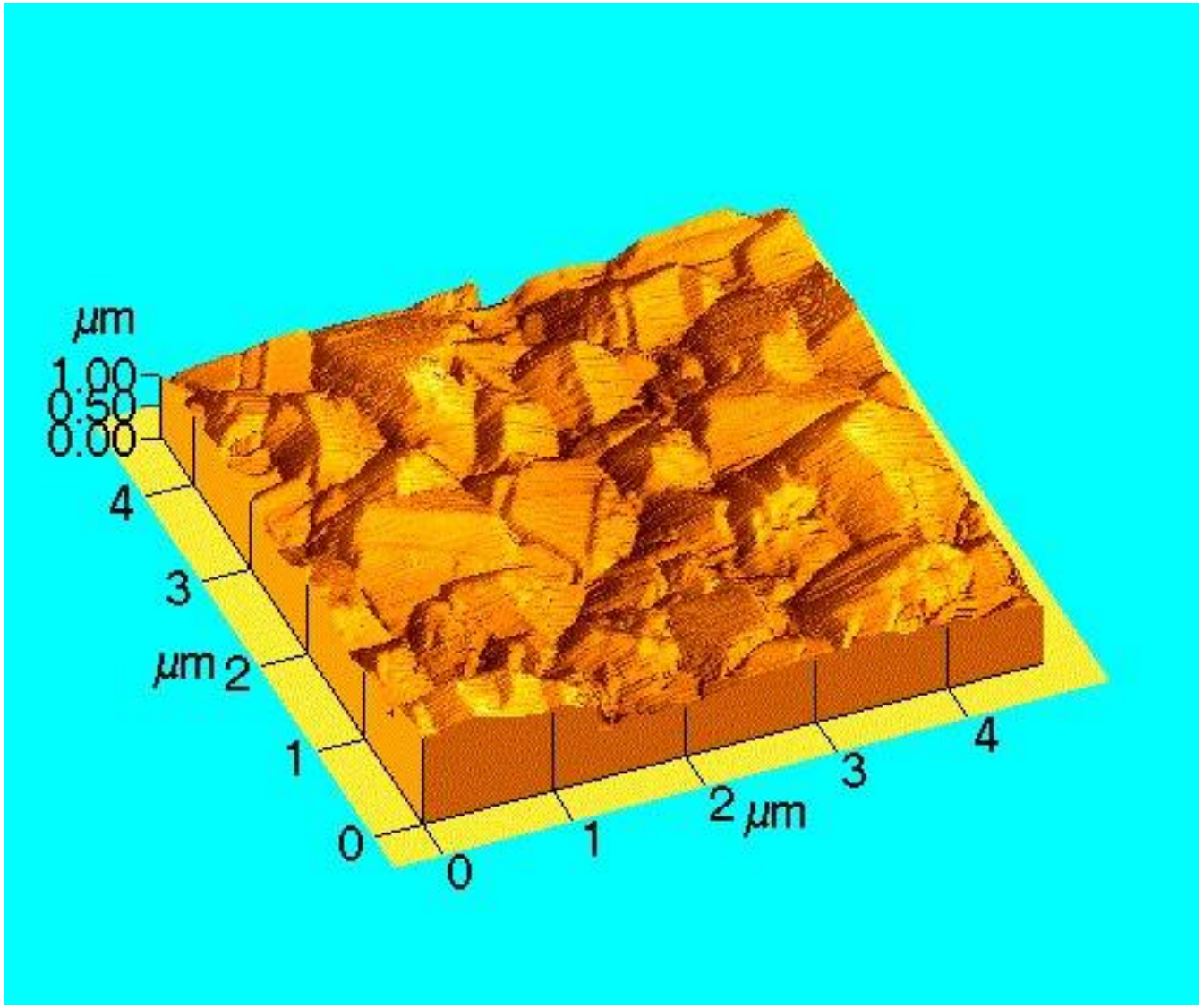
Scanning Probe Microscope (SPM)

SPMs are used for studying surface topography and properties of materials from the *atomic* to the *micron* level.



↓ to pography

SPM



Microscopy

Question:

→ they don't depend on pcs ←

Why were commercial TEMs developed from about 1938 and SEMs from about 1965, whereas SPMs were not around before 1980's?

↳ depends on pc

Answer:

TEMs and SEMs were not limited by computers technology initially as images could be easily collected on photographic film which has been around for a long time, with high resolution. On the other hand, SPMs ***depend*** *on computers to construct the 3D images.*

General Resolution of Microscopes

Type of Microscope	Approx. Resolution
Human Eye	2'500'000 Å
Optical Light (OLM)	3000 Å
Scanning Electron (SEM)	10-50 Å
Transmission Electron (TEM)	2-5 Å, near atomic
Scanning Probe (SPM)	0.1 Å, atomic

Review Electron Microscopy

A fracture surface is very rough, with peaks and valleys. Why does the SEM give a clear image of it while the optical microscope gives an image where only part is in focus?

in slide 14

A. The SEM has a much larger depth of field.

B. The SEM has higher magnification.

C. The SEM uses colour.

D. The fracture surface must be etched first.

Use the QR code shown in the classroom. This one won't work.

Next topic:

Diffusion